Lab Performance Test

Student ID: Course Code: **CSE-336** Duration: **40 minutes**

Instructions:

1. Create four source files named 2204XXX_A.sh, 2204XXX_B.sh, 2204XXX_C.sh, 2204XXX_D.sh. (XXX is equal to your student id).
2. Upload your solution program in the google classroom.
3. **Do not use internet (Browser, Whatsapp, Chatgpt, etc) during 1 Hour exam. If we find this, you will be expelled from this test**

Problem A:

Problem Description:

Write a shell script that accepts an integer input from the user and evaluates its divisibility properties with respect to the numbers 3 and 5. The program should determine whether the given number is divisible by both 3 and 5, divisible only by 3, divisible only by 5, or not divisible by either of them.The script must implement appropriate **conditional statements** to perform the required checks and display a clear, user-friendly message indicating the result of the evaluation. The script should handle negative values correctly and be structured using simple if-else statements.

Sample Input Output:

Sample Input	Sample Output
15	Divisible by both 3 and 5
9	Divisible by 3 only
7	Not divisible by 3 or 5

Problem B:

Problem Description:

Write a shell script that takes a traffic light color as input **(red, yellow, green)** and prints the correspondi action using a **case** statement. The script must follow the action below:

Color	Action
Red	Stop
Green	Go
Yellow	Get Ready
Otherwise	Invalid Signal

Sample Input Output:

Sample Input	Sample Output
Red	Stop
Black	Invalid Signal
green	Go

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Problem C:

Problem Description:

Write a shell script that calculates the factorial of a number **N** using a while loop. **[Note that the program will input the number N as command line arguments]**

Sample Input Output:

Sample Input	Sample Output
2204XXX_C.sh 5	120
2204XXX_C.sh 8	40320
2204XXX_C.sh 1	1

Problem D:

Problem Description:

Write a recursive function using shell script to calculate **X raised to the power Y (X^Y). [Note that program will input the numbers X and Y as command line arguments]**

Sample Input Output:

Sample Input	Sample Output
2204XXX_D.sh 2 4	16
2204XXX_D.sh 5 3	125
2204XXX_D.sh 7 2	49

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